

Magnesium

Fact Sheet for Health Professionals

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Introduction

Magnesium, an abundant mineral in the body, is naturally present in many foods, added to other food products, available as a dietary supplement, and present in some medicines (e.g., antacids, laxatives). Magnesium is a cofactor in more than 300 enzyme systems that regulate diverse biochemical reactions in the body, including protein synthesis, muscle and nerve function, blood glucose control, and blood pressure regulation [1-3]. Magnesium is required for energy production, oxidative phosphorylation, and glycolysis. It contributes to the structural development of bone and is required for the synthesis of DNA, RNA, and the antioxidant glutathione. Magnesium also plays a role in the active transport of calcium and potassium ions across cell membranes, a process that is important to nerve impulse conduction, muscle contraction, and normal heart rhythm [3].

An adult body contains approximately 25 grams magnesium, with 50% to 60% present in the bones and most of the rest in soft tissues [4]. Less than 1% of total magnesium is in blood serum, and these levels are kept under tight control. Normal serum magnesium concentrations range between 0.75 and 0.95 millimoles/liter (mmol/L) [1,5]. Hypomagnesemia is defined as a serum magnesium level less than 0.75 mmol/L [6]. Magnesium homeostasis is largely controlled by

the kidneys, which typically excrete about 120 milligrams (mg) magnesium into the urine each day [2]. Urinary excretion is reduced when magnesium status is low [1].

Assessing magnesium status

Because most of the magnesium in the body is found inside cells or in bone, it is difficult to assess magnesium status. Measuring serum magnesium concentration is the most commonly used method for assessment, but serum levels do not accurately reflect total body magnesium levels or concentrations in specific tissues. Comprehensively evaluating magnesium status may require the use of both laboratory tests and a clinical assessment.

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Recommended Intakes

The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for magnesium. These values range from 310 to 420 mg for adults and from 30 to 410 mg for infants, children, and adolescents, depending on age, sex, and life stage.

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Sources of Magnesium

Food

Good sources of magnesium include green leafy vegetables, legumes, nuts, seeds, whole grains, and certain beverages. Magnesium may also be added to some breakfast cereals and other fortified foods. In general, approximately 30% to 40% of the magnesium obtained from food and beverages is absorbed by the body.

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Dietary supplements

Supplements can contain a variety of different forms of magnesium, and the absorption of these forms varies. In general, forms of magnesium that dissolve well in liquid have higher absorption than other forms, and the aspartate, citrate, lactate, and chloride forms of magnesium tend to have higher bioavailability than magnesium oxide and magnesium sulfate.

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Medicines

Magnesium is found in some remedies for heartburn and acid indigestion, and it is a key ingredient in certain laxatives.

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Magnesium Intakes and Status

Many people in the United States consume less than the recommended amounts of magnesium, with men age 71 years and older and adolescent males and females having the highest risk of low magnesium intakes. Survey data suggest that the average intakes of magnesium from food alone are higher among people who take dietary supplements that contain magnesium than among those who do not. There is no current data on magnesium status among people who live in the United States, as magnesium status is not routinely evaluated in national surveys, hospitals, or clinics.

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Magnesium Deficiency

Loss of appetite, nausea, vomiting, fatigue, and weakness are some of the early signs and symptoms of magnesium deficiency. As the condition progresses, people with magnesium deficiency may experience numbness, muscle contractions and cramps, seizures, abnormal heart rhythms, coronary spasms, and other symptoms. In severe cases, magnesium deficiency can cause hypocalcemia or hypokalemia. While certain health conditions and the use of certain medications can cause symptomatic magnesium deficiency, it is uncommon in healthy people.

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Groups at Risk of Magnesium Inadequacy

Certain groups of people are more likely than others to have magnesium inadequacy. These include people with gastrointestinal diseases, type 2 diabetes, or alcohol dependence and older adults.

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Magnesium and Health

Habitually low intakes of magnesium induce changes in biochemical pathways that can increase the risk of illness over time. This section focuses on four diseases and disorders in which magnesium might be involved: hypertension and cardiovascular disease, type 2 diabetes, osteoporosis, and migraine headaches.

Hypertension and cardiovascular disease

Results from clinical trials suggest that magnesium supplementation only marginally lowers blood pressure. Although some prospective studies have reported that people with higher intakes and/or serum levels of magnesium have lower risks of sudden cardiac death, ischemic heart disease, and stroke, more research is needed to understand the effects of magnesium from food and dietary supplements on cardiovascular disease.

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Type 2 diabetes

Several meta-analyses of prospective cohort studies have reported that individuals who have higher magnesium intakes tend to have a lower risk of type 2 diabetes. However, only a few small clinical trials have evaluated using supplemental magnesium to control type 2 diabetes, and these trials have reported conflicting results. According to the American Diabetes Association, there is currently not enough evidence to recommend using magnesium to improve glycemic control in people with diabetes.

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Osteoporosis

Magnesium is essential for bone health, and some studies have found that higher intakes of magnesium are associated with increased bone mineral density. Other studies have reported finding low serum magnesium levels among women with osteoporosis. However, more research is needed to determine whether magnesium supplements can help prevent or manage osteoporosis.

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Migraine headaches

Low levels of serum and tissue magnesium appear to be associated with the occurrence of migraine headaches. Some research has suggested that the regular use of magnesium supplements may reduce the frequency of migraines. Although the data are limited, the American Academy of Neurology and the American Headache Society have concluded that magnesium is probably effective for migraine prevention.

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Health Risks from Excessive Magnesium

Although the risk of acquiring too much magnesium from food is low among healthy people, high doses of magnesium from dietary supplements or medications can cause diarrhea, nausea, and abdominal cramping. Very high doses of magnesium can lead to magnesium toxicity, which can cause hypotension, vomiting, difficulty breathing, irregular heartbeat, cardiac arrest, and other signs and symptoms. A few cases of fatal hypermagnesemia have been reported. The Tolerable Upper Intake Level for supplemental magnesium is 350 mg for adults, and it ranges from 65 to 350 mg for children and adolescents, depending on age.

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Interactions with Medications

Magnesium may interact with certain medications, such as oral bisphosphonates, tetracyclines, and quinolone antibiotics. In addition, some medications, including diuretics and proton pump inhibitors, can affect magnesium status.

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Magnesium and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. *The Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

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